

Saim Bashir

Software Engineering Undergraduate | Mobile Engineering | Applied AI | AI-Assisted Software Engineering

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2 Google Play releases | 3 mobile products | 7 student workflows | 10+ gameplay systems

PROFILE

- Built and shipped **2 Android applications** on Google Play and currently have a third mobile product in closed testing.
- Comfortable moving from an idea to architecture, implementation, debugging, device testing, and release preparation instead of working on only one part of the product.
- Use AI-assisted engineering workflows for architecture planning, code review, bug investigation, test design, and documentation, then validate suggested changes through code inspection, builds, and device tests.

EDUCATION

BS Software Engineering, COMSATS University Islamabad, Lahore Campus Feb 2025 - Jan 2029 (Expected)

- **CGPA: 3.58/4.00** | Fourth semester.
- **Core computing:** Object-Oriented Programming, Data Structures, Data Analysis & Algorithms, Software Engineering, Database Systems, Computer Networks, Artificial Intelligence with Python.
- **Math foundation:** Discrete Structures, Statistics & Probability, Linear Algebra, Calculus and Analytical Geometry.
- Coursework is reinforced through independent Android products, algorithm-focused assignments, database work, and iterative testing on real devices.

TECHNICAL SKILLS

Programming	C++, Java, JavaScript, Python, SQL
Mobile & APIs	React Native, Expo, Android development, REST API integration , local persistence, notifications, responsive UI
Software Engineering	OOP, version control , modular architecture, code review, debugging, testing, release workflows
Algorithms & Data	Linked lists, stacks, queues, trees, heaps, graphs, hash maps, BFS, DFS, searching, sorting, recursion, complexity analysis
Developer Tooling	Git, GitHub, Visual Studio Code, MySQL, Android build and device-testing workflows
AI / Agent Workflow	Claude Code, OpenCode , reusable Agent Skills , architecture planning, code review, refactoring support, bug investigation, test planning, technical documentation

SELECTED ENGINEERING PROJECTS

University Timetable Pro | Published Android Application React Native, Expo

Student productivity, attendance logic, notifications, cognitive mini-games

Google Play: play.google.com/store/apps/details?id=com.saimbashir.universitytimetable

- Built one app around **7 everyday student workflows**, covering schedules, attendance, remaining absences, lecture reminders, alerts, and activity planning instead of spreading them across separate tools.
- Added **3 short cognitive challenges** for memory, logic, and reflexes, plus daily streaks, so productive planning and short study breaks live in the same product.
- Shipped the app to Google Play after working through local data, notifications, Android device behavior, release builds, privacy preparation, and repeated real-device testing.
- Used **Claude Code, OpenCode**, and reusable **Agent Skills** to accelerate architecture planning, code review, bug investigation, and test design. Proposed changes were checked in code and validated through builds or device tests before integration.

Starlight Catcher | Published Android Game React Native, Expo

Game systems, collision logic, progression design, release engineering

Google Play: play.google.com/store/apps/details?id=com.saimbashir.neoncolorcatcher

- Turned a semester Java OOP game into a touch-first Android product and carried it through the extra engineering needed for real users and store distribution.
- Designed and integrated **10+ gameplay systems**, including themed levels, target matching, unlockables, coins, power-ups, hazards, combo rewards, daily rewards, audio, visual effects, and progressive difficulty.
- Repeatedly tested movement, collisions, scoring, boundaries, saved rewards, and device behavior before completing the second Google Play release.

Pattern Pulse | Cognitive Pattern Game

React Native, Expo | Closed Testing

Memory, focus, pattern recognition, engagement systems

- Building a cognitive game around short visual-pattern challenges that become harder as the player progresses and rely on memory, attention, recognition, and quick response.
- Combined **5 engagement systems**, including levels, scoring, coins, performance history, and daily rewards, with a bonus milestone every **20 completed levels**.
- Closed testing is being used to tune difficulty, reward clarity, startup stability, visual consistency, and the final Android release experience.

Recall Lens AI | Multimodal Personal Knowledge Retrieval

Python | RAG | Vector Search | Completed

Semantic retrieval, multimodal memory search, grounded question answering, knowledge connections

- Developed an AI-assisted personal knowledge retrieval system for PDFs, screenshots, images, notes, voice clips, and saved resources using OCR, speech-to-text, semantic embeddings, vector search, retrieval-augmented generation (RAG), and grounded question answering.
- Implemented timeline-aware search, smart summaries, knowledge connections, and privacy controls; storage and data structures include vector embedding indexes, SQLite relational tables, knowledge-graph nodes/edges, and metadata/tag mappings.

ACADEMIC ENGINEERING PROJECTS

Grocery Management System

C++ | Data Structures

Linked structures, record operations, memory management

- Built the system with **3 linked data structures**: linked lists, stacks, and queues, choosing each structure for the operation it handled best.
- Implemented **5 core record operations**: add, search, update, process, and remove, while working directly with pointers, traversal, insertion, deletion, and LIFO/FIFO behavior.

Starlight Catcher | OOP Prototype

Java | Object-Oriented Programming

Game entities, events, collisions, score and life state

- Used **6 core OOP and interaction concepts**: classes and objects, encapsulation, inheritance, polymorphism, event handling, and collision logic.
- Added score, lives, levels, visual feedback, and increasing difficulty. This prototype later became the starting point for the released React Native version.

Cafe Management System

C++ | Programming Fundamentals

Console workflow, validation, structured problem solving

- Built a complete console workflow for menu selection, orders, quantities, and billing using functions, arrays, loops, conditionals, input validation, and structured decomposition.

RESEARCH READINESS

- Start from an observed user or software problem, break it into requirements and edge cases, then test whether the implemented behavior actually solves it.
- Compare alternatives in areas such as reminder behavior, interaction sensitivity, reward timing, level progression, and interface layout before settling on a final design.
- Use **version control**, repeatable builds, structured debugging, test planning, and documentation so changes can be traced, reviewed, and reproduced.
- Looking to deepen formal research skills in literature review, experimental design, user studies, algorithm comparison, data analysis, and technical writing.

RESEARCH-RELEVANT TECHNICAL BACKGROUND

- **LLM code generation and refactoring**: Practical fit through OOP, modular architecture, version control, code review, debugging, test planning, and hands-on AI-assisted development workflows.
- **Educational systems and adaptive interfaces**: Timetable Pro and Pattern Pulse already connect mobile UI, cognitive activities, engagement systems, user feedback, and repeated interface refinement.
- **Data analytics and machine learning**: Coursework in AI with Python, statistics, linear algebra, SQL, algorithms, and databases provides the academic base for data-oriented research projects.
- **Complex networks and anomaly detection**: Graphs, BFS, DFS, searching, sorting, complexity analysis, Python, and structured debugging provide a useful starting point for learning ML-based anomaly detection on network data.

ADDITIONAL ACADEMIC INFORMATION

Intermediate in Computer Science, KIPS College: 997/1200 (83.08%)

Matriculation, OPF Public High School: 1054/1100 (95.82%)

Languages: Urdu (Native), English (Working academic proficiency)

Summer 2027 availability: 06/18/27 - 09/20/27